Vectors

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Vectors

Exercise 0.1 (Store a data point as one object)**.** A house has a floor area of 120 square meters, 3 bedrooms, and an age of 15 years.

1. Write these three numbers as a single vertical list $\tilde{x}$, in the order given, and call its entries x_1, x_2 and x_3. What is x_2?
2. Does the list $(3, 120, 15)$ describe the same house?

Solution

Solution 0.1.

1. Stacking the three numbers in order gives

$$\tilde{x} = \begin{bmatrix} 120 \\ 3 \\ 15 \end{bmatrix},$$

   so $x_2 = 3$, the number of bedrooms.

2. No. The list $(3, 120, 15)$ would describe a house with 3 square meters of floor area and 120 bedrooms. An entry's position says what it measures, so the order of the entries matters.

Definition 0.1 (Column vector)**.** A **column vector** of length p is an ordered list of p numbers, written vertically:

$$\tilde{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix}$$

Remark (Vectors are columns by default)

Remark 0.1 (Vectors are columns by default). In these notes, as in most statistics textbooks, a vector is a column vector unless stated otherwise. A vector written inline as a list in parentheses is still a column vector: for example, $\tilde{x} = (2, -1, 5)$ means

$$\tilde{x} = \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}.$$

Definition 0.2 (The set $\mathbb{R}^p$)**.** For a positive integer p, $\mathbb{R}^p$ is the set of all column vectors of length p whose entries are real numbers.

Remark (Reading $\tilde{x} \in \mathbb{R}^p$)

Remark 0.2 (Reading $\tilde{x} \in \mathbb{R}^p$). Writing $\tilde{x} \in \mathbb{R}^p$ says, in one symbol, that $\tilde{x}$ is a column vector with p real entries. For example, $(1, -2, 0.5) \in \mathbb{R}^3$. The vector $(1, 2)$ is not in $\mathbb{R}^3$, because it has only 2 entries; it is in $\mathbb{R}^2$. In Exercise 0.1, $\tilde{x} \in \mathbb{R}^3$.

Definition 0.3 (Scalar)**.** In linear algebra, a **scalar** is a single real number, as opposed to a vector or a matrix, which holds several numbers.

Example 0.1 (Scalars and a vector)**.** The numbers 5 and -0.2 are scalars. The column vector $(5, -0.2)$ is not a scalar: it has 2 entries, so it is in $\mathbb{R}^2$ (Definition 0.2), and each of its 2 entries is a scalar.

Definition 0.4 (Real-valued (scalar-valued) function)**.** A function $f : A \to \mathbb{R}$, whose values are numbers, is **real-valued** (also called **scalar-valued**, since its values are scalars).

Definition 0.5 (Vector-valued function)**.** A function $f : A \to \mathbb{R}^m$, whose values are vectors (Definition 0.2), is **vector-valued**.

Example 0.2 (A real-valued and a vector-valued function)**.** On $\mathbb{R}^2$, $f(\tilde{x}) = x_1 + x_2$ is real-valued and $g(\tilde{x}) = \begin{bmatrix} x_1 + x_2 \\ x_1 - x_2 \end{bmatrix}$ is vector-valued:

$$f\left(\begin{bmatrix} 3 \\ 4 \end{bmatrix}\right) = 3 + 4 = 7, \qquad g\left(\begin{bmatrix} 3 \\ 4 \end{bmatrix}\right) = \begin{bmatrix} 3 + 4 \\ 3 - 4 \end{bmatrix} = \begin{bmatrix} 7 \\ -1 \end{bmatrix}$$

Definition 0.6 (Transpose)**.** The **transpose** of a column vector $\tilde{x}$ is the row vector with the same sequence of entries, written horizontally:

$$\tilde{x}^\top \equiv \tilde{x}' \equiv [x_1,\ x_2,\ \ldots,\ x_p]$$

Example 0.3 (Transposing a column vector)**.** The transpose of the column vector with entries 2, −1, 5 is the row vector with the same entries:

$$\begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}^\top = [2,\ -1,\ 5].$$

Definition 0.7 (Vector addition)**.** The **sum** of two column vectors $\tilde{x}$ and $\tilde{y}$ of the same length p is the column vector $\tilde{x} + \tilde{y}$ of length p obtained by adding entry by entry:

$$(\tilde{x} + \tilde{y})_i \stackrel{\text{def}}{=} x_i + y_i, \quad i = 1, \ldots, p$$

Example 0.4 (Adding two vectors)**.**

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 + 4 \\ 2 + 5 \\ 3 + 6 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix}$$

Definition 0.8 (Dot product)**.** For any two real-valued vectors $\tilde{x} = (x_1, \ldots, x_p)$ and $\tilde{y} = (y_1, \ldots, y_p)$ of the same length p, the **dot product** of $\tilde{x}$ and $\tilde{y}$ is:

$$\tilde{x} \cdot \tilde{y} = \tilde{x}^\top \tilde{y} \stackrel{\text{def}}{=} \sum_{i=1}^{p} x_i y_i \tag{1}$$

See also the definitions in:

- Dobson and Barnett (2018), Section 1.3 (equation 1.1, page 7)
- Kaplan (2022), chapter on vectors
- wikipedia

The dot product has a different generalization for two matrices; see wikipedia for more.

Example 0.5 (A dot product)**.** For $\tilde{x} = (1, 2, 3)$ and $\tilde{y} = (4, 5, 6)$:

$$\begin{aligned} \tilde{x} \cdot \tilde{y} &= 1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6 && \text{(definition of the dot product)} \\ &= 4 + 10 + 18 && \text{(multiply)} \\ &= 32 && \text{(add)} \end{aligned}$$

Definition 0.9 (Linear combination)**.** A **linear combination** of the numbers $a_1, \ldots, a_k$ with coefficients $c_1, \ldots, c_k$ (Definition 0.10) is the weighted sum

$$c_1 a_1 + \cdots + c_k a_k = \sum_{i=1}^{k} c_i a_i$$

A linear combination of vectors $\tilde{v}_1, \ldots, \tilde{v}_k$ of the same length is defined entry by entry: entry j of $\sum_{i=1}^{k} c_i \tilde{v}_i$ is the linear combination of the jth entries of $\tilde{v}_1, \ldots, \tilde{v}_k$ with the same coefficients.

Definition 0.10 (Coefficients of a linear combination)**.** The numbers $c_1, \ldots, c_k$ in a linear combination (Definition 0.9) $\sum_{i=1}^{k} c_i a_i$ are its **coefficients**.

Example 0.6 (Coefficients of a linear combination of numbers)**.** The linear combination of the numbers 4 and 5 with coefficients 2 and 3 is $2 \cdot 4 + 3 \cdot 5 = 8 + 15 = 23$.

Example 0.7 (A linear combination of two vectors)**.** The linear combination of $(1, 0)^\top$ and $(0, 1)^\top$ with coefficients 2 and 3 is $(2 \cdot 1 + 3 \cdot 0,\ 2 \cdot 0 + 3 \cdot 1)^\top = (2, 3)^\top$.

Remark (The dot product is a linear combination)

Remark 0.3 (The dot product is a linear combination). The dot product $\tilde{x} \cdot \tilde{y}$ is a linear combination (Definition 0.9) of the entries of $\tilde{y}$, with coefficients $x_1, \ldots, x_p$. In Example 0.5, $\tilde{x} \cdot \tilde{y} = 1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6$ is the linear combination of 4, 5, and 6 with coefficients 1, 2, and 3.

The dot product is also the standard *inner product* on $\mathbb{R}^p$; "inner product" is the general notion (Definition 1 in Inner Products and Orthogonality, later in these notes), of which the dot product is one example.

Theorem 0.1 (Dot product is symmetric)**.** *The dot product is symmetric:*

$$\tilde{x} \cdot \tilde{y} = \tilde{y} \cdot \tilde{x}$$

Proof

Proof.

$$\begin{aligned} \tilde{x} \cdot \tilde{y} &= \sum_{i=1}^{p} x_i y_i && \text{(definition of the dot product)} \\ &= \sum_{i=1}^{p} y_i x_i && \text{(products of numbers are symmetric)} \\ &= \tilde{y} \cdot \tilde{x} && \text{(definition of the dot product)} \end{aligned}$$

□

Special vectors

Definition 0.11 (Zero vector)**.** The **zero vector** $\tilde{0}$ of length p has all entries equal to zero:

$$\tilde{0} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Remark (The zero vector is the additive identity)

Remark 0.4 (The zero vector is the additive identity). Adding the zero vector to a vector leaves it unchanged (Definition 0.7): $\tilde{x} + \tilde{0} = \tilde{x}$ for any vector $\tilde{x}$ of the same length, so $\tilde{0}$ is the identity element for vector addition. For example,

$$\begin{aligned} (2, -1) + (0, 0) &= (2 + 0, -1 + 0) \\ &= (2, -1). \end{aligned}$$

Definition 0.12 (Mean (average, sample mean))**.** The **mean** (or **average**) of n numbers $y_1, \ldots, y_n$ is their sum divided by how many there are:

$$\bar{y} \stackrel{\text{def}}{=} \frac{1}{n} \sum_{i=1}^{n} y_i$$

Example 0.8 (The mean of three numbers)**.** For $y_1 = 2$, $y_2 = -1$ and $y_3 = 5$:

$$\begin{aligned} \bar{y} &= \frac{1}{3}(2 + (-1) + 5) && \text{(definition of the mean, with } n = 3) \\ &= \frac{1}{3} \cdot 6 && \text{(add)} \\ &= 2 && \text{(multiply)} \end{aligned}$$

Remark (The sample mean estimates the population mean)

Remark 0.5 (The sample mean estimates the population mean). When the numbers are observed data values, $\bar{y}$ is also called the sample mean, an estimator of the population mean.

Definition 0.13 (Ones vector)**.** The **ones vector** $\tilde{1}$ of length p has all entries equal to one:

$$\tilde{1} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

Remark (The ones vector sums the entries)

Remark 0.6 (The ones vector sums the entries). The dot product (Definition 0.8) of $\tilde{1}$ with $\tilde{x}$ is the sum of the entries of $\tilde{x}$:

$$\begin{aligned} \tilde{1} \cdot \tilde{x} &= \sum_{i=1}^{p} 1 \cdot x_i \\ &= \sum_{i=1}^{p} x_i. \end{aligned}$$

For example, for $\tilde{x} = (2, -1, 5)$,

$$\begin{aligned}\tilde{1} \cdot \tilde{x} &= 2 + (-1) + 5 \\ &= 6,\end{aligned}$$

and dividing by the length $p = 3$ gives the mean of the entries, $6/3 = 2$.

Definition 0.14 (Indicator vector (standard basis vector))**.** The j-th **indicator vector** (or **standard basis vector**) $\tilde{e}_j$ of length p has a 1 in position j and 0s elsewhere:

$$(\tilde{e}_j)_i = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \qquad \tilde{e}_j = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \leftarrow \text{position } j$$

Remark (Indicator vectors build every vector)

Remark 0.7 (Indicator vectors build every vector). For $p = 3$, the indicator vectors are $\tilde{e}_1 = (1, 0, 0)$, $\tilde{e}_2 = (0, 1, 0)$, and $\tilde{e}_3 = (0, 0, 1)$. Every vector of length 3 is a weighted sum $c_1\tilde{e}_1 + c_2\tilde{e}_2 + c_3\tilde{e}_3$ of them, with weights equal to its entries; for example, $(2, -1, 5) = 2\tilde{e}_1 - 1\tilde{e}_2 + 5\tilde{e}_3$. No other weights work: entry j of $c_1\tilde{e}_1 + c_2\tilde{e}_2 + c_3\tilde{e}_3$ is c_j, so each weight c_j must equal entry j of the vector.

Theorem 0.2 (Indicator vectors select entries)**.** *For any vector $\tilde{x}$ of length p and any $j \in \{1, \ldots, p\}$:*

$$\tilde{e}_j \cdot \tilde{x} = x_j$$

Proof

Proof. Writing the dot product componentwise:

$$\begin{aligned}\tilde{e}_j \cdot \tilde{x} &= \sum_{i=1}^{p} (\tilde{e}_j)_i \, x_i && \text{(definition of the dot product)} \\ &= \sum_{i=1}^{p} \begin{cases} 1 \cdot x_i & \text{if } i = j \\ 0 \cdot x_i & \text{if } i \neq j \end{cases} && \text{(definition of } \tilde{e}_j\text{)} \\ &= x_j && \text{(only the } i = j \text{ term is nonzero)}\end{aligned}$$

□

Theorem 0.3 (A vector is a linear combination of indicator vectors)**.** *Every vector $\tilde{x} \in \mathbb{R}^p$ is the linear combination (Definition 0.9) of the indicator vectors $\tilde{e}_1, \dots, \tilde{e}_p$ (Definition 0.14) with coefficients $x_1, \dots, x_p$:*

$$\tilde{x} = \sum_{j=1}^{p} x_j \tilde{e}_j$$

Proof

Proof. Compare the two sides entry by entry. For each $i \in \{1, \dots, p\}$:

$$\begin{aligned}
\left(\sum_{j=1}^{p} x_j \tilde{e}_j\right)_i &= \sum_{j=1}^{p} x_j (\tilde{e}_j)_i && \text{(Definition 0.9, entry by entry)} \\
&= \sum_{j=1}^{p} \begin{cases} x_j \cdot 1 & \text{if } j = i \\ x_j \cdot 0 & \text{if } j \neq i \end{cases} && \text{(Definition 0.14)} \\
&= x_i \cdot 1 + \sum_{j \neq i} x_j \cdot 0 && \text{(split off the } j = i \text{ term)} \\
&= x_i + 0 && \text{(times 1 changes nothing; times 0 gives 0)} \\
&= x_i && \text{(adding 0 changes nothing)}
\end{aligned}$$

□

Example 0.9 (Expanding a vector in indicator vectors)**.**

$$\begin{aligned}
\begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix} &= 4 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + (-1) \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\
&= 4\tilde{e}_1 - \tilde{e}_2 + 2\tilde{e}_3
\end{aligned}$$

Orthogonality

Definition 0.15 (Orthogonal vectors)**.** Two vectors $\tilde{x}$ and $\tilde{y}$ of the same length are **orthogonal** (written $\tilde{x} \perp \tilde{y}$) if their dot product (Definition 0.8) is zero:

$$\tilde{x} \perp \tilde{y} \iff \tilde{x} \cdot \tilde{y} = 0$$

Remark (Orthogonal means perpendicular)

Remark 0.8 (Orthogonal means perpendicular). Orthogonality extends the geometric idea of perpendicular lines to vectors with any number of entries. In the plane, $(1, 2)$ and $(-2, 1)$ are perpendicular, and their dot product is $1 \cdot (-2) + 2 \cdot 1 = 0$. The same test works in $\mathbb{R}^3$, where pictures are harder to draw:

$$\begin{aligned}(1, 1, 0) \cdot (1, -1, 5) &= 1 - 1 + 0 \\ &= 0,\end{aligned}$$

so $(1, 1, 0) \perp (1, -1, 5)$.

Example 0.10 (Vectors that are not orthogonal)**.**

$$\begin{aligned}(1, 2) \cdot (1, 1) &= 1 \cdot 1 + 2 \cdot 1 \\ &= 3 \\ &\neq 0,\end{aligned}$$

so $(1, 2)$ and $(1, 1)$ are not orthogonal.

Definition 0.16 (Euclidean norm (length, L_2 norm))**.** The **Euclidean norm** (or **length**, also called the L_2 **norm** and written $\|\tilde{x}\|_2$) of a vector $\tilde{x}$ of length p is

$$\|\tilde{x}\| \stackrel{\text{def}}{=} \sqrt{\tilde{x} \cdot \tilde{x}} = \sqrt{\sum_{i=1}^{p} x_i^2} \tag{2}$$

where $\sqrt{\cdot}$ is the square root.

Example 0.11 (The length of a vector)**.** For $\tilde{x} = (3, 4)$:

$$\begin{aligned} \|\tilde{x}\| &= \sqrt{3^2 + 4^2} && \text{(definition of the norm)} \\ &= \sqrt{25} && \text{(square and add)} \\ &= 5 \end{aligned}$$

The vector $(0.6, 0.8)$ has norm $\sqrt{0.36 + 0.64} = 1$.

Definition 0.17 (Unit vector)**.** A **unit vector** is a vector whose Euclidean norm (Definition 0.16) is 1. A unit vector is said to have *unit length*.

Example 0.12 (A unit vector and a vector that is not one)**.** For $\tilde{x} = (0.6, 0.8)$:

$$\begin{aligned} \|\tilde{x}\| &= \sqrt{0.6^2 + 0.8^2} && \text{(definition of the norm)} \\ &= \sqrt{0.36 + 0.64} && \text{(square)} \\ &= \sqrt{1} && \text{(add)} \\ &= 1 && \text{(take the square root)} \end{aligned}$$

so $(0.6, 0.8)$ is a unit vector. The vector $(1, 1)$ is not:

$$\begin{aligned} \|(1,1)\| &= \sqrt{1^2 + 1^2} \\ &= \sqrt{2} \\ &\neq 1. \end{aligned}$$

Exercise 0.2 (How far apart are two points?)**.** Let $\tilde{x} = (1, 2)$ and $\tilde{y} = (4, 6)$.

1. Compute the vector $\tilde{x} - \tilde{y}$.
2. Compute the length of $\tilde{x} - \tilde{y}$, using the Euclidean norm.
3. Is the result the same if you compute the length of $\tilde{y} - \tilde{x}$ instead?

Solution

Solution 0.2.

1.
$$\begin{aligned} \tilde{x} - \tilde{y} &= (1 - 4, 2 - 6) \\ &= (-3, -4). \end{aligned}$$

2.
$$\begin{aligned}\|\tilde{x}-\tilde{y}\| &= \sqrt{(-3)^2+(-4)^2} \\ &= \sqrt{25} \\ &= 5.\end{aligned}$$

3. $\tilde{y}-\tilde{x}=(3,4)$, and
$$\begin{aligned}\|\tilde{y}-\tilde{x}\| &= \sqrt{3^2+4^2} \\ &= 5.\end{aligned}$$

Yes, the result is the same.

Definition 0.18 (Euclidean distance)**.** The **Euclidean distance** between two vectors $\tilde{x}$ and $\tilde{y}$ of length p is the Euclidean norm of their difference:
$$d(\tilde{x},\tilde{y}) \stackrel{\text{def}}{=} \|\tilde{x}-\tilde{y}\| = \sqrt{\sum_{i=1}^{p}(x_i-y_i)^2} \tag{3}$$

Remark (Distance from the origin)

Remark 0.9 (Distance from the origin). The distance from the origin $\tilde{0}$ to a vector $\tilde{x}$ is the norm of $\tilde{x}$:
$$\begin{aligned}d(\tilde{x},\tilde{0}) &= \left\|\tilde{x}-\tilde{0}\right\| \\ &= \|\tilde{x}\|\end{aligned}$$
(Goodfellow et al. 2016, 39).
For example, the distance from $\tilde{0}$ to $(3,4)$ is $\|(3,4)\|=5$, as in Example 0.11.

Definition 0.19 (Orthonormal vectors)**.** A set of vectors $\{\tilde{x}_1,\tilde{x}_2,\ldots,\tilde{x}_k\}$ is **orthonormal** if the vectors are mutually orthogonal (Definition 0.15) and each has norm 1 (Definition 0.16), that is, unit length:
$$\tilde{x}_i\cdot\tilde{x}_j = \begin{cases}1 & \text{if } i=j \\ 0 & \text{if } i\neq j\end{cases}$$

Example 0.13 (Indicator vectors are orthonormal)**.** The indicator vectors $\tilde{e}_1, \tilde{e}_2, \ldots, \tilde{e}_p$ (Definition 0.14) form an orthonormal set. By Theorem 0.2, $\tilde{e}_i \cdot \tilde{e}_j$ is entry i of $\tilde{e}_j$, which is 1 if $i = j$ and 0 if $i \neq j$. For $p = 2$:

$$\begin{aligned} \tilde{e}_1 \cdot \tilde{e}_1 &= 1 \cdot 1 + 0 \cdot 0 \\ &= 1, \end{aligned}$$

$$\begin{aligned} \tilde{e}_2 \cdot \tilde{e}_2 &= 0 \cdot 0 + 1 \cdot 1 \\ &= 1, \end{aligned}$$

and

$$\begin{aligned} \tilde{e}_1 \cdot \tilde{e}_2 &= 1 \cdot 0 + 0 \cdot 1 \\ &= 0. \end{aligned}$$

Example 0.14 (Two ways to fail to be orthonormal)**.**

- $(1, 1)$ and $(1, -1)$ are orthogonal, since $1 \cdot 1 + 1 \cdot (-1) = 0$, but not orthonormal: $(1, 1) \cdot (1, 1) = 2 \neq 1$.
- $(1, 0)$ and $(0.6, 0.8)$ both have unit length, since $(1, 0) \cdot (1, 0) = 1$ and

  $$\begin{aligned} (0.6, 0.8) \cdot (0.6, 0.8) &= 0.36 + 0.64 \\ &= 1, \end{aligned}$$

  but they are not orthonormal: $(1, 0) \cdot (0.6, 0.8) = 0.6 \neq 0$.

Exercise 0.3 (Compute an inner product and a norm)**.** Let

$$\tilde{a} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}, \qquad \tilde{b} = \begin{bmatrix} 0 \\ 4 \\ -2 \end{bmatrix}$$

Compute $\tilde{a}^\top \tilde{b}$ and $\|\tilde{a}\|_2$.

Solution

Solution 0.3. Multiply entry by entry and add:

$$\begin{aligned}\tilde{a} \cdot \tilde{b} &= (3)(0) + (-1)(4) + (2)(-2) \\ &= 0 - 4 - 4 \\ &= -8\end{aligned}$$

For the norm, take the inner product of $\tilde{a}$ with itself first:

$$\begin{aligned}\tilde{a} \cdot \tilde{a} &= 3^2 + (-1)^2 + 2^2 \\ &= 9 + 1 + 4 \\ &= 14\end{aligned}$$

so

$$\|\tilde{a}\|_2 = \sqrt{14} \approx 3.742$$

The inner product came out negative while the norm cannot: a norm is a square root of a sum of squares, so it is never below zero.
Python has both operations built in, and R builds them from `sum`.

```python
import numpy as np

a = np.array([3.0, -1.0, 2.0])
b = np.array([0.0, 4.0, -2.0])

print(a @ b, round(float(np.linalg.norm(a)), 3))
```

```r
a <- c(3, -1, 2)
b <- c(0, 4, -2)

print(c(sum(a * b), round(sqrt(sum(a^2)), 3)))
#> [1] -8.000  3.742
```

Both print -8 and 3.742.

Video lecture

Hutchinson's Linear Function Basics (24 min) covers dot products and the Euclidean norm, and goes on to hyperplanes (Definition 22 in Matrices) and level sets (Hutchinson, n.d.). The login for the video site is posted on Canvas.

Dobson, Annette J, and Adrian G Barnett. 2018. *An Introduction to Generalized Linear Models*. 4th ed. CRC press. https://doi.org/10.1201/9781315182780.

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